

Exam: Crosstabs & Contingency Analysis

Practice Exam with Solutions | Basic Level

NOTES

- Enter your name and student ID on the cover sheet.
- Time: 60 minutes. Total points: 30 pts.
- Permitted aids: Formula sheet, calculator, pen.
- Application tasks: Document solution steps. Clearly mark your results.
- Solutions are provided in the appendix for self-assessment.

QUESTIONS

1. [Application] (6 Pkt.) In a crosstab with variables 'Gender' (m/f) and 'Driver license' (yes/no), the total is $n = 200$. Out of 100 men, 80 have a license; out of 100 women, 60 have one.

- a) Create the 2×2 crosstab with absolute frequencies and margins. (2 pts.)
- b) Calculate row percentages. What do they indicate? (2 pts.)
- c) Calculate column percentages. What do they indicate? (2 pts.)

2. [Open question] (3 Pkt.) When do you use row percentages, column percentages, and total percentages? Briefly explain with a self-chosen example.

3. [Application] (5 Pkt.) A study examines the relationship between vaccination status (vaccinated/unvaccinated) and illness (yes/no). Absolute frequencies:

Vaccinated: 20 ill, 180 healthy. Unvaccinated: 50 ill, 50 healthy.

Calculate:

- a) Relative risk (RR) of falling ill for unvaccinated vs. vaccinated (2 pts.)
- b) Odds ratio (OR) (2 pts.)
- c) Interpret both values (1 pt.)

4. [Open question] (3 Pkt.) What is Simpson's Paradox? Explain with an example why a single crosstab can be misleading.

5. [Application] (5 Pkt.) A researcher surveys $n = 400$: Field of study (STEM/Languages/Social) and study abroad (yes/no).

Results: STEM 80 Yes / 60 No; Languages 50 Yes / 70 No; Social 70 Yes / 70 No.

- a) Calculate column percentages. (2 pts.)
- b) Conduct a conceptual chi-square test. At $\chi^2 = 9.84$ and $\alpha = 0.05$: What is the decision? ($\chi^2_{\text{crit}}(2) = 5.99$) (2 pts.)
- c) Calculate Cramér's V. (1 pt.)

SOLUTIONS

1.a) Table:

License	Yes	No	Total
Male	80	20	100
Female	60	40	100
Total	140	60	200

b) Row %: Male 80% Yes, 20% No. Female 60% Yes, 40% No.

Indicates: A higher proportion of men have a license (80% vs 60%).

c) Column %: License Yes: 57.1% m, 42.9% f. No: 33.3% m, 66.7% f.

Indicates: Among license holders, 57% are male.

2. Row %: IV in rows compare DV across row categories.

Column %: IV in columns compare DV across column categories.

Total %: Purely descriptive overview without causal direction.

Example: 'Education (IV) × Voting (DV)' row %, because education (row) is the assumed cause.

3. Table:

Ill	Healthy	Total	
Vaccinated	20	180	200
Unvaccinated	50	50	100
Total	70	230	300

a) $RR = (50/100) / (20/200) = 0.5 / 0.1 = 5.0$.

Unvaccinated persons have a 5-fold risk of falling ill.

b) $OR = (50 \cdot 180) / (20 \cdot 50) = 9000 / 1000 = 9.0$.

Odds of illness are 9 times higher for the unvaccinated.

c) Both indicate a strong association. $OR > RR$ because OR overestimates with frequent events. RR is more intuitive.

4. Simpson's Paradox: The relationship between two variables reverses when a third confounding variable is controlled for.

Example (Berkeley): Overall, men are admitted at a higher rate. Within each department, admission is balanced. Reason: women apply more to departments with lower admission rates.

Solution: Stratified tables (Mantel-Haenszel) or logistic regression.

5.a) Column % (column = abroad yes/no):

Abroad Yes: STEM 40%, Languages 25%, Social 35%

Abroad No: STEM 30%, Languages 35%, Social 35%

STEM students are overrepresented among those who studied abroad.

b) $\chi^2 = 9.84 > 5.99$ reject H_0 . Significant relationship between field of study and study abroad ($p < 0.05$).

c) $V = \sqrt{9.84 / (400 \cdot 1)} = \sqrt{0.0246} = 0.157$ small effect.